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Optical Coherence Tomography Angiography

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Introduction

Optical coherence tomography (OCT) is a noninvasive imaging technique that uses low-coherence interferometry to produce depth-resolved imaging. A beam of light is used to scan an eye area, say the retina or anterior eye, and interferometrical measurements are obtained by interfering with the backscatter or reflectance from ocular structures with the known reference path of traveling light.[1] This modification of classic Michelson interferometry allows using OCT to generate structural anatomy images.[2] OCT has become widely adopted in ophthalmology since its introduction in 1991 and has continually improved.[1][3] Until optical coherence tomography angiography (OCTA), conventional structural OCT images predominantly provided visualization of anatomic changes with low contrast between small blood vessels and tissue within retinal layers. Thus, other imaging modalities, such as fluorescein or indocyanine green angiography, were generally used to evaluate retinal vasculature and choroidal vasculature, respectively.[4][5]

OCTA uses the principle of diffractive particle movement of moving red blood cells to determine vessel location through various segments of the eye without the need for any intravascular dyes.[6] OCTA technology allows for the ability to image flow in the retinal and choroidal vasculature through en-face, depth-encoded slabs. These slabs are presented alongside structural OCT B-scans, which obtain cross-sectional images. Together, they provide detailed flow imaging of the deep retinal vascular plexus and choriocapillaris, which were poorly visualized with previous imaging modalities.

Distinguishing the differences between Doppler OCT and OCTA is critical. Although they both use phase information, Doppler OCT quantifies blood flow in larger vessels and measures total retinal blood flow using phase shift. In contrast, OCTA analyzes scatter from a static background tissue to create angiograms.[7][8]

Historical Development of Optical Coherence Tomography

OCT is the optical equivalent of ultrasound, which generates images using time delay and light echo magnitudes. Michel Duguay originally proposed using echoes of light to examine biological tissue at AT&T Bell Laboratories when he published "Light Photographed in Flight" in American Scientist in 1971 and was the first to show that high-speed shutters made it possible to "see inside biological tissues." [9]

The field of femtosecond optics was further developed by Erich Ippen of the Massachusetts Institute of Technology (MIT) in the mid-1970s. His group collaborated with Dr. Carmen Puliafito of the Massachusetts Eye and Ear Infirmary; together, they studied femtosecond laser effects on the retina and the cornea. Moreover, working with DeSilvestri of Milan, Italy, and Margolis and Oseroff from the Department of Dermatology at Massachusetts General Hospital, Duguay's initial work to "see inside tissues" was further developed.[10] These investigators initially used lasers at a 625-nm wavelength and later progressed to using longer 1300-nm wavelengths, which allowed the

reduction of scattering. The first application of low-coherence interferometry, which was used to measure the eye's axial length, was reported by Fercher et al of the Medical University of Vienna, Austria, in 1988.[11]

An electrical engineering undergraduate, John Apostolopoulos, used low-coherence laser diodes in 1989 to describe the potential ophthalmic applications of this technology, although the sensitivity was limited. However, a significant breakthrough was made through the ongoing research into low-coherence interferometry of David Huang, an MD/PhD student, in 1991. Huang showed the practical applicability of coherence interferometry using an 800-nm low-coherence laser diode. Consequently, higher sensitivities were achieved, which yielded information on eye structures (eg, the lens and the iris). The first OCT images were published by Huang in Science in 1991. Unpublished concepts of a similar system were also shown by Tanno et al in Japan.

Swanson et al developed the first in vivo retinal images in 1993, and Fercher et al of Vienna demonstrated a similar retinal system. Practical advances were then rapidly made by the MIT group working with Carmen Puliafito and Joel Schuman of the New England Eye Center of the Tufts University School of Medicine in Boston. OCT examination protocols for circumpapillary scanning for the assessment of glaucoma and macular edema were developed by Michael Hee, an expert programmer who used the early Apple Macintosh computers.[12]

Michael Hee was largely responsible for major developments in the 1990s, publishing more than 30 papers during his doctorate. His 1997 doctorate thesis, "Optical Coherence Tomography of the Eye," remains a seminal reference work on OCT in ophthalmology. The first OCT atlas was organized by Carmen Puliafito in 1996 (*Optical Coherence Tomography of Ocular Diseases*, Slack, 1995). The Advanced Ophthalmic Diagnostics company, set up by C Puliafito, E Swanson, and J Fujimoto in 1992, was acquired by Humphrey Zeiss 2 years later and went on to develop machines that were introduced into clinical use, the first machine being introduced in 1996.

As with many new techniques, clinical adoption by the ophthalmic community was slow in the latter 1990s, with only 180 units in use until 2000. By 2004, the company had developed faster machines with better-resolution images; by 2004, more than 10 million OCT imaging procedures had been obtained worldwide. OCT has since become a standard of care in the ophthalmic community. OCT imaging is now used in various subspecialties, including ophthalmology, cardiovascular medicine, dermatology, neurology, gastroenterology, dentistry, otolaryngology, urology, pulmonology, and gynecology, with new applications found every year.

Specimen Collection

Conventional OCT devices obtain an axial scan or A-scan by analyzing the reflectance or scatter of light of various tissue structures at different depths. A B-scan, a cross-sectional image, is then generated by combining multiple A-scans obtained as the light beam scans the tissue in the transverse direction. Acquiring multiple B-scans displaced perpendicular to the B-scan image can provide volumetric information that composes a raster scan. The elements expanded upon through OCTA are supplied by the motion of blood within the tissue, which is predominantly the only movement when scanning the stationary retina. Thus, comparing repeated OCT B-scans allows blood flow imaging by comparing the pixel-by-pixel differences between scans.[2]

OCTA allows the noninvasive blood flow detection and 3D representation of vasculature. OCTA identifies vasculature by analyzing differences within a repeatedly scanned transverse cross-sectional area of tissue. Current methods for motion detection include amplitude-decorrelation and phase variance. Amplitude decorrelations work by detecting the difference in amplitude between OCT B-scans, whereas phase variance compares the variations of the emitted light wave properties when it intercepts moving objects.[13] Both methods utilize the concept that nonmobile tissue will remain identical on consecutive OCT scans, whereas moving erythrocytes will cause changes in successive OCT scans. Furthermore, OCTA uses the same idea between spectral-domain OCT, wavelengths near 800 nm, and swept-source OCT, wavelengths near 1050 nm, where longer wavelengths have high penetrance to analyze vasculature at various tissue levels.

Due to the high sensitivity to movement during OCTA imaging, devices commonly implement eye-tracking methods to improve visualization and reduce background interference. The most notable methods are the split spectrum amplitude decorrelation and volume averaging techniques.[14][15]

Procedures

OCTA is a cutting-edge, noninvasive imaging technique that allows the visualization of retinal and choroidal microvasculature in vivo. Unlike traditional fluorescein angiography, OCTA does not require dye injections, making it a safer and faster alternative with fewer risks of adverse reactions.

The OCTA procedure is quick, safe, and invaluable in diagnosing and monitoring retinal and choroidal diseases. This procedure's noninvasive nature and high-resolution imaging capabilities make it an essential tool in modern ophthalmic care. Therefore, proper acquisition, image analysis, and clinical interpretation are crucial to maximizing OCTA's diagnostic potential.[16][17] The following are essential components of OCTA:

- **Preprocedure preparation:** Patient education and preparation are essential before conducting OCTA. The patient should be informed about the procedure, including the noninvasive nature of the test and the absence of any dye injections. The patient is seated in front of the OCTA machine, and proper alignment is crucial to ensure high-quality images. Pupillary dilation may be required for certain patients, particularly if the optical media are unclear.[18]
- **Patient positioning:** Proper positioning is essential to obtain optimal scans. The patient is instructed to rest their chin on the chinrest and forehead against the headband. The machine will automatically adjust to the patient's position to ensure the eye is centered on the camera. The patient is asked to maintain steady fixation on the target to minimize motion artifacts during scanning.[19]
- **Acquisition of images:** OCTA uses motion contrast to detect blood flow in the retinal and choroidal vessels. It captures repeated B-scans (cross-sectional scans) at the exact location, and flow information is extracted by detecting changes between consecutive scans. This process allows for the creation of detailed en-face images of the microvasculature. The scanning process takes only a few seconds. OCTA typically captures scans at different retina and choroid layers, including the superficial vascular plexus, deep vascular plexus, outer retina, and choriocapillaris. The machine also automatically segments the retina into these layers for further analysis. [2]
- **Segmentation and layering:** The OCTA software automatically segments the retina and choroid into different layers following image acquisition. Manual correction may be required if the automatic segmentation misidentifies layers, particularly in patients with pathologies, eg, diabetic retinopathy or macular degeneration. [8]
 - **Superficial retinal layer:** captures the vasculature within the inner retina
 - **Deep retinal layer:** provides information about the deeper retinal vasculature
 - **Outer retina layer:** normally avascular, but abnormalities in neovascular diseases can be detected
 - **Choriocapillaris:** shows blood flow within the choriocapillaris and choroid [8]
- **Image analysis:** A thorough analysis is performed once the images are obtained. The OCTA software provides various analysis tools clinicians utilize to evaluate the vasculature for signs of pathology, including neovascularization, nonperfusion, microaneurysms, or vascular abnormalities, particularly in diabetic retinopathy, macular degeneration, and retinal vein occlusions.

- **Flow density maps:** highlight areas of blood flow, allowing for comparison between healthy and diseased regions
- **Three-dimensional reconstruction:** created by some devices, 3D vasculature models provide a more comprehensive view of the retina and choroid
- **En-face imaging:** software-generated en-face images to assess specific vascular layers in the retina and choroid [8]
- **Postprocedure care:** No specific postprocedure care is required after the OCTA procedure. Patients can resume their normal activities immediately, as no dilation or invasive techniques are involved. If pupil dilation was used for the procedure, patients may be advised to avoid driving until the effects wear off.[20]
- **Clinical interpretation:** Clinicians review OCTA images to detect retinal or choroidal circulation abnormalities. Some of the key features analyzed include:
 - **Capillary dropout or nonperfusion:** seen in diabetic retinopathy or vein occlusions
 - **Choroidal neovascularization:** associated with age-related macular degeneration and other conditions
 - **Microaneurysms or retinal telangiectasia:** detected in diabetic patients or other vascular conditions
 - **Optic nerve head perfusion:** essential in evaluating glaucoma patients [21]
- **Follow-up:** Based on the OCTA findings, the clinician may recommend follow-up imaging to monitor disease progression or treatment response. For example, in cases of diabetic retinopathy or macular degeneration, repeated scans may be scheduled to assess the impact of interventions like anti-VEGF therapy.[17]

Indications

The clinical indications for using OCTA have not been determined by the field as a whole. Still, OCTA capabilities open a vast array of possibilities for research into the pathogenesis of diseases, disease quantification, and treatment evaluation, including:

- Isolate locations of vascular pathology
- Analyze each retinal vascular plexus separately
- Evaluate the foveal avascular zone
- Examine the perifoveal endocapillary area
- Examine retinal microcirculation
- Evaluate response to anti-VEGF therapy [3][6][22]

Optical Coherence Tomography Angiography Versus Fluorescein Angiography and Indocyanine Green Angiography

Fluorescein angiography and indocyanine green angiography (ICGA) require the administration of intravenous dye. Fluorescein angiography has been used in clinical practice for over 50 years and is the gold standard for detecting retinal neovascularization, disc neovascularization, and choroidal neovascularization. Both fluorescein angiography and ICGA supply a 2D image with visualization of dynamic blood flow, including patterns of dye leakage, pooling of dye, and staining of structures.[23] Although OCTA cannot show the dynamic properties of dye within the vasculature, it does not require injectable dye, allows for quantitative and qualitative analysis of multiple layers, and provides 3D and 2D imaging.[24]

OCTA has rapidly gained prominence as a noninvasive imaging modality in ophthalmology, offering high-resolution visualization of blood flow in the retina and choroid without needing dye injection. The primary indications for OCTA include:

- **Diabetic retinopathy:** OCTA is indicated for the early detection and monitoring of microvascular changes in diabetic patients, including capillary non-perfusion and microaneurysms, even before the appearance of clinical symptoms.[25]
- **Age-related macular degeneration:** OCTA is critical in detecting and monitoring choroidal neovascularization in dry and wet age-related macular degeneration (AMD), allowing timely intervention to prevent vision loss. [26]
- **Retinal vein occlusion:** OCTA helps identify areas of capillary nonperfusion, macular edema, and the development of collateral vessels in cases of branch and central retinal vein occlusion.[27]
- **Glaucoma:** OCTA is indicated for assessing microvascular changes in the optic nerve head and peripapillary region, helping in the early diagnosis and monitoring of glaucoma progression.[28]
- **Macular telangiectasia:** OCTA visualizes abnormal telangiectatic vessels in the macula, aiding in diagnosing and managing macular telangiectasia.[29]
- **Central serous chorioretinopathy:** OCTA helps detect choroidal vascular changes and subretinal fluid in patients with CSCR, providing insight into disease progression and treatment efficacy.[30]
- **Uveitis and retinal vasculitis:** OCTA is valuable in detecting retinal vasculitis and macular edema in inflammatory conditions, guiding treatment decisions.[31]
- **Myopic choroidal neovascularization:** High myopia patients benefit from OCTA for early detection of choroidal neovascularization, which can develop as a complication of pathological myopia.[32]
- **Hereditary retinal dystrophies:** Inherited retinal conditions such as retinitis pigmentosa can be monitored using OCTA to assess vascular changes, particularly capillary dropout and retinal atrophy.[33]
- **Ocular ischemic syndrome:** OCTA is indicated for identifying ischemic changes in the retinal and choroidal vasculature in patients with carotid artery disease or other vascular disorders affecting ocular blood flow.[34]
- **Coats disease:** In pediatric and rare vascular conditions like Coats disease, OCTA is used to visualize abnormal retinal vasculature and detect associated exudates.[35]
- **Idiopathic polypoidal choroidal vasculopathy:** OCTA helps diagnose and monitor polypoidal lesions in the choroid, which can lead to subretinal hemorrhage and fluid accumulation.[36]
- **Retinal and choroidal tumors:** OCTA can assist in assessing the vascular network in ocular tumors, helping to differentiate between benign and malignant lesions.[37]
- **Postsurgical monitoring:** OCTA is indicated for postsurgical evaluation of retinal and choroidal blood flow, particularly after procedures such as vitrectomy or retinal laser treatments.[38]

These indications demonstrate the broad utility of OCTA across a spectrum of retinal and choroidal diseases, making it a vital tool for diagnosis and management in ophthalmic practice.

Potential Diagnosis

OCTA is a revolutionary imaging modality allowing noninvasive, high-resolution visualization of blood flow within the retina and choroid. Due to its detailed imaging capabilities, OCTA has become a valuable diagnostic tool for various retinal and choroidal diseases. The ability of OCTA to visualize vascular changes noninvasively makes it particularly useful in early detection and assessing disease progression. By identifying normal and pathological blood flow patterns, OCTA aids in diagnosing conditions, eg, diabetic retinopathy, age-related macular degeneration, and glaucoma, while offering valuable insights into rarer conditions like macular telangiectasia and inherited retinal dystrophies.[39]

Diabetic Retinopathy

OCTA plays a crucial role in the early detection and management of diabetic retinopathy, even before clinical symptoms manifest. Diabetic retinopathy findings that may be demonstrated with OCTA include:

- **Capillary nonperfusion areas:** Loss of blood flow in retinal capillaries is a hallmark of diabetic retinopathy.
- **Microaneurysms:** The lesions appear as small, hyperreflective lesions on OCTA, indicating localized vascular swelling.
- **Neovascularization:** New abnormal blood vessels can be visualized as irregular formations breaking through the normal retinal layers. OCTA clearly assesses these vessels, which are prone to bleeding and leakage.
- **Foveal avascular zone enlargement:** Diabetic retinopathy can cause an irregular foveal avascular zone (FAZ) enlargement, signaling early macular ischemia and potential vision loss.[40]

Age-Related Macular Degeneration

OCTA is particularly useful for detecting choroidal neovascularization (CNV) associated with wet AMD. Key findings in AMD patients using OCTA include:

- **Subretinal neovascular membranes:** New vessels originating from the choroid penetrate the retinal pigment epithelium and cause fluid leakage, leading to vision loss.
- **Drusen:** Dry AMD is characterized by small deposits under the retina that may disrupt blood flow. By detecting early neovascular changes, OCTA helps monitor the transition from dry to wet AMD.
- **FAZ alterations:** In late AMD, the FAZ can become irregular or enlarged due to ischemic processes.[41]

Retinal Vein Occlusion

OCTA helps visualize the extent of vascular compromise in retinal vein occlusion (RVO). Findings characteristic of RVO include:

- **Capillary nonperfusion:** Blood flow disruptions in the superficial and deep capillary plexuses indicate ischemia due to vein occlusion.
- **Collateral vessels:** The formation of new vessels to bypass the blocked veins can be visualized.
- **Macular edema:** OCTA can demonstrate fluid buildup as hyporefective spaces in the macular region, often seen in branch RVO and central RVO.[42]

Glaucoma

In glaucoma, OCTA can identify microvascular damage in the optic nerve head and peripapillary regions, which is essential for early diagnosis and monitoring of disease progression. Findings characteristic of glaucoma include:

- **Decreased perfusion in the optic nerve head:** Reduced vascular density around the optic nerve correlates with damage to retinal ganglion cells.
- **Capillary dropout:** Loss of capillaries in the superficial and deep retinal layers is a sign of advanced glaucoma. [43]

Macular Telangiectasia

In this condition, OCTA can detect telangiectatic vessels in the macular region, which often appear abnormal and can cause fluid leakage, leading to macular edema. Characteristic findings include:

- **Vascular dilation:** Abnormal widening of the macular capillaries.
- **Foveal vascular anomalies:** The FAZ may be irregular, with vessels encroaching on the central macula, leading to visual disturbances. [29]

Central Serous Chorioretinopathy

OCTA is effective in visualizing abnormalities in the choroidal vasculature associated with central serous chorioretinopathy (CSCR), including:

- **Choroidal vascular hyperpermeability:** Blood flow disturbances in the choriocapillaris can be detected, aiding in the diagnosis of CSCR.
- **Subretinal fluid accumulation:** OCTA can visualize fluid collections in the macular area without invasive dye studies, helping monitor disease progression. [30]

Uveitis and Vasculitis

In patients with inflammatory conditions affecting the eye, OCTA can reveal the following:

- **Retinal vasculitis:** Vessel wall thickening and irregular blood flow patterns indicate inflammation in the retinal vasculature.
- **Macular edema:** OCTA can demonstrate cystoid spaces and fluid accumulation, a common complication in uveitis.
- **Choroidal involvement:** In some uveitic conditions, the choroid can become inflamed, leading to alterations in the choriocapillaris flow. [31]

Hereditary Retinal Dystrophies

OCTA can be used in inherited retinal conditions such as retinitis pigmentosa and cone-rod dystrophies to detect the following:

- **Reduced vascular density:** OCTA can show progressive capillary loss in these diseases, particularly in the outer retina and choroid.
- **Foveal vascular alterations:** The foveal vasculature may become irregular or atrophied as the disease progresses, leading to central vision loss. [33]

Myopia-Associated Pathology

High myopia is associated with several retinal and choroidal pathologies that can be visualized using OCTA. Characteristic findings include:

- **Choroidal thinning:** OCTA can quantify the thinning of the choriocapillaris, which correlates with myopic degeneration.
- **Choroidal neovascularization detection:** Choroidal neovascularization can develop in pathologic myopia, and OCTA helps detect these neovascular lesions early.[32]

Coats Disease

In this rare condition, characterized by abnormal retinal vessel development and leakage, OCTA can identify the following:

- **Vascular malformations:** OCTA can visualize abnormal telangiectatic vessels and aneurysms within the retina.
- **Subretinal exudates:** Fluid accumulation beneath the retina due to vascular leakage can be monitored.[44]

Normal and Critical Findings

OCTA is a noninvasive imaging modality that allows visualization of the retinal and choroidal vasculature. It provides high-resolution images of blood flow without the need for dye injection, as required in traditional fluorescein angiography. OCTA is used to diagnose and manage various retinal and choroidal diseases. Understanding normal and critical findings in OCTA is crucial for accurate interpretation and treatment planning.[45]

Normal Findings

In a healthy individual, OCTA reveals a well-defined vasculature in the superficial and deep retinal layers and the choroidal circulation. The key features of a normal OCTA scan include the following:

- **Superficial retinal capillary plexus:** This plexus lies between the inner limiting membrane and the inner plexiform layer. The vessels are evenly spaced and organized, with a uniform flow pattern.[46]
- **Deep retinal capillary plexus:** Located deeper within the retina, between the inner plexiform layer and outer plexiform layer (OPL), the DCP is denser and more complex, displaying a network of microvascular structures. [47]
- **Avascular zone:** The FAZ is a critical finding in normal eyes. It is a round, well-defined region in the center of the macula with no blood vessels. The FAZ is typically tiny and circular in healthy individuals, contributing to high visual acuity.[48]
- **Choriocapillaris:** Below the retina, the choriocapillaris appear as a uniform layer of blood flow. This layer supplies oxygen and nutrients to the outer retina.[49]
- **Vascular flow:** In normal findings, the flow signals should be continuous and smooth across all retinal and choroidal layers without any signs of disruption or blockage.[50]

Critical Findings

OCTA can also reveal critical pathological changes in patients with various ocular conditions. Recognizing these findings helps clinicians assess disease severity and progression, which include:

- **Nonperfusion areas:** In conditions like diabetic retinopathy or retinal vein occlusion, OCTA may show areas of capillary nonperfusion where blood flow is absent. These areas appear as dark zones within the vasculature, indicating ischemia and an increased risk of neovascularization.[51]
- **Vessel dropout:** Capillary dropout or loss is visible in glaucoma or vascular diseases, particularly in the deep retinal layers. This is associated with the progression of disease and visual field loss.[52]

- **Foveal avascular zone enlargement:** Enlargement or irregularity of the FAZ is a critical finding in macular diseases such as diabetic macular edema or macular telangiectasia. A larger FAZ correlates with poorer visual outcomes.[53]
- **Neovascularization:** OCTA is highly sensitive to detecting neovascularization, eg, AMD. Abnormal blood vessels that breach the retinal pigment epithelium or proliferate in the retinal layers are considered critical. These new vessels are often irregular and leaky, posing a risk for severe vision loss.[54]
- **Cystoid spaces:** In diseases retinal vein occlusion or uveitis, cystoid spaces (cystoid macular edema) may appear on OCTA as hyporeflective spaces within the retina, disrupting normal blood flow.[55]
- **Vascular abnormalities:** Inherited retinal diseases such as macular telangiectasia or Coats disease show aberrant vascular patterns, with dilation or telangiectatic vessels visible on OCTA.[56]
- **Flow voids in the choriocapillaris:** OCTA of the choriocapillaris may reveal areas of flow voids, which could indicate pathology (eg, choroidal neovascularization or inflammatory diseases).[57]
- **Reduced flow signal:** Flow signals may be significantly reduced in conditions like retinal artery occlusion or high myopia, particularly in the deeper vascular layers.[58]

Clinical Relevance

Understanding normal and critical findings in OCTA is essential for diagnosing age-related macular degeneration, diabetic retinopathy, retinal vein occlusions, and glaucoma. OCTA also aids in assessing treatment response and disease progression. By comparing regular baseline scans with pathological changes, OCTA enables early detection and intervention, thereby improving patient outcomes. Ongoing studies [39] are examining normal and abnormal findings that may be associated with certain conditions observable by OCTA, including:

- **Diabetes**
 - Choriocapillaris abnormalities
 - Retinal microvasculature abnormalities
 - Vascular remodeling of adjacent foveal avascular zone
 - Enlarged foveal avascular zone
 - Capillary tortuosity and dilation
 - Areas of retinal nonperfusion
 - Reduced capillary density [59]
- **Dry age-related macular degeneration**
 - Areas of impaired choriocapillaris flow extending beyond areas of geographic atrophy
 - A decrease in choriocapillaris density [60]
- **Neovascular age-related macular degeneration**
 - Choriocapillaris alterations surrounding choroidal neovascularization
 - Retinal angiomatous proliferation [61]
- **Vascular occlusion**
 - Areas of capillary nonperfusion with clear delineation of the ischemic boundary

- Microaneurysms
- Telangiectasis
- Anastomosis [42]
- **Glaucoma**
 - Attenuation of peripapillary microvasculature [62]

Interfering Factors

OCTA is not without its limitations. OCTA is highly sensitive to motion, and although eye-tracking methods are commonly deployed within devices, patient collaboration is required.[63] Motion artifacts can appear as white or black lines or misalign the retinal vasculature. Segmentation errors can also occur when imaging an abnormal retina, although it can often be manually edited.[64]

OCTA technology is also dependent on the light source and, as such, is limited by media opacities, leading to signal attenuation and shadowing artifacts.[13] This aspect can also be seen when superficial blood vessels obscure deeper vessels.[24] Additionally, a projection artifact can occur when superficial vasculature erroneously appears in segmented views of deeper layers.[65][62] Lastly, the current automated area of visualizations in OCTA spans from 2 mm to 12 mm, which can also add the limitation of being unable to image the peripheral retina.[62][65]

Interfering factors concerning OCTA can affect imaging quality and its diagnostic utility. These factors must be understood to minimize their impact and optimize OCTA's clinical utility. The following are some key interfering factors:

- **Small field of view:** OCTA provides a relatively limited field of view compared to traditional wide-field fluorescein angiography. This limitation can interfere with detecting peripheral retinal changes, especially in diabetic retinopathy.[66]
- **False positives and negatives:** Due to high sensitivity, OCTA may detect nonpathological variations or fail to capture slow-flow vessels, leading to diagnostic errors.[67]
- **Variability between devices:** Different OCTA machines may have variability in imaging algorithms and hardware, affecting the consistency of results across devices.[68]
- **Patient cooperation:** Cooperation is essential for achieving high-quality imaging. Anxiety, poor fixation, or inability to remain still can negatively impact scan quality.[69]

Understanding these factors helps clinicians achieve accurate results, minimize errors, and enhance patient outcomes when utilizing OCTA in clinical practice.[67]

Complications

OCTA is a widely utilized, noninvasive imaging technology that visualizes the retina and choroid microvascular structures. Although OCTA offers many benefits in diagnosing and monitoring various ocular conditions, this study has limitations and potential complications.[17] While the risks are far fewer than with traditional angiography techniques (eg, fluorescein angiography), understanding the following complications that may arise is crucial for clinicians and patients alike:

- **Motion artifacts**
 - **Description:** One of the most common complications of OCTA is motion artifacts caused by involuntary eye movements or blinking during the scan. These artifacts can create distorted or blurred images,

making it difficult to interpret vascular details accurately.

- **Impact:** Artifacts can lead to misdiagnosis or failure to detect subtle vascular changes, especially in chronic diseases like diabetic retinopathy or AMD.[70]

- **Shadowing artifacts**

- **Description:** Shadowing artifacts occur when large blood vessels or dense ocular media (eg, cataracts) block the OCT signal, causing shadows that obscure underlying structures.
- **Impact:** This can limit the visibility of deeper retinal or choroidal vessels, especially in patients with advanced disease or opacities in the eye, leading to incomplete assessments.[71]

- **Limited depth penetration**

- **Description:** OCTA has limited depth penetration, which restricts its ability to capture deeper retinal and choroidal structures, particularly in cases with significant media opacity or other obstacles.
- **Impact:** This limitation may result in an incomplete diagnosis or misinterpretation, especially in patients with conditions affecting the deeper retinal or choroidal layers.[72]

- **False positives or negatives**

- **Description:** Due to its high sensitivity, OCTA may detect artifacts or nonpathological variations in vascular structures, leading to false positives. Conversely, certain slow-flow or microvascular abnormalities may not be detected, resulting in false negatives.
- **Impact:** Misinterpretation due to false positives or negatives can delay appropriate treatment or lead to unnecessary interventions, increasing patient anxiety or risk.[65]

- **Inability to detect leakage**

- **Description:** OCTA is excellent for visualizing blood flow but does not detect vascular leakage, a key feature in certain retinal conditions like macular edema.
- **Impact:** This limitation means that OCTA often needs to be supplemented with traditional fluorescein angiography in cases where detecting leakage is clinically essential for diagnosis and treatment planning. [73]

- **Poor image quality in media opacity**

- **Description:** Patients with significant cataracts or corneal opacities may not achieve optimal imaging with OCTA, as the clarity of the scan is compromised.
- **Impact:** This could lead to insufficient or low-quality images, limiting the clinical utility of the test for these patients and necessitating alternative imaging methods.[74]

- **Limitations in capturing large fields of view**

- **Description:** OCTA captures smaller fields of view compared to wide-field fluorescein angiography.
- **Impact:** This may result in the inability to detect peripheral vascular abnormalities, which are common in diseases like proliferative diabetic retinopathy.[75]

- **Difficulty with segmentation**

- **Description:** Accurate segmentation of retinal and choroidal layers is essential for analyzing OCTA data, but automatic segmentation algorithms may fail in cases of severe pathology, such as large macular scars

or highly irregular ocular anatomy.

- **Impact:** Incorrect segmentation could lead to misdiagnosis or prevent accurate analysis of the vascular structures in the retina or choroid.[76]

- **Interpretation complexity**

- **Description:** OCTA produces highly detailed images that require advanced interpretation skills. Variations in normal anatomy, as well as artifacts, may make interpretation difficult, particularly for clinicians who are not experienced in using OCTA.
- **Impact:** Inaccurate interpretation can lead to unnecessary interventions or delays in the correct diagnosis and treatment.[65]

- **Variability in image quality across devices**

- **Description:** Significant variability is present in the image quality produced by different OCTA devices. This variability can be related to differences in scanning protocols, signal processing algorithms, and imaging depth.
- **Impact:** Lack of standardization across devices may complicate comparing images over time or between different clinics, potentially leading to inconsistencies in monitoring disease progression.[77]

- **Patient discomfort or anxiety**

- **Description:** Although OCTA is a noninvasive procedure, some patients may experience discomfort due to the bright lights used during imaging. Additionally, anxious patients may find it challenging to stay still during the scan, leading to suboptimal results.
- **Impact:** Poor cooperation can lead to motion artifacts requiring repeat scans, which may prolong the procedure and increase patient anxiety.[17]

- **Costs and accessibility**

- **Description:** While not a complication in the clinical sense, the cost of OCTA technology and lack of widespread access may pose challenges in specific healthcare settings.
- **Impact:** Patients may face financial constraints, or clinicians may not have access to this advanced technology, which could limit the availability of OCTA for some populations.[78]

- **Overuse or misapplication**

- **Description:** A risk of overusing OCTA in situations where traditional methods could suffice may arise. Clinicians may also inappropriately apply OCTA findings without correlating them with clinical presentations.
- **Impact:** This could lead to overdiagnosis or overtreatment and increased patient healthcare costs.[39]

While OCTA is a revolutionary tool in modern ophthalmology, clinicians must know its potential complications and limitations. Understanding these challenges is critical for obtaining accurate images, making correct diagnoses, and optimizing patient outcomes. Additionally, educating patients about the potential limitations of OCTA can help manage expectations and improve cooperation during the procedure. Ultimately, OCTA should be used judiciously, supplemented with other diagnostic tools when necessary, and interpreted within the broader context of each patient's clinical condition.[79]

Patient Safety and Education

OCT is a noninvasive imaging technology used to obtain high-resolution cross-sectional images of the retina. The layers within the retina can be differentiated, and retinal thickness can be measured for early detection and diagnosis of retinal diseases. Patients may be given an OCT scan to monitor the progression of the disease, verify suspected swelling in the retina, or evaluate the adverse effects of medication. OCT testing is a standard of care for diagnosing, managing, and treating most retinal conditions. The OCT uses light rays to measure retinal thickness and does not exhibit radiation or x-rays. No eye contact is involved during the OCT scan, and generally, the procedure is painless and well-tolerated when the patient is positioned correctly. OCT is commonly compared to ultrasound, except it uses light rather than sound.[80]

Patients may be administered dilating eye drops to widen the pupil for imaging. Typically, the patient sits in front of the OCT machine while the patient's head rests on a support. The patient must remain still and follow the instructions of the imager when directed to look in a specific direction. Scanning the eye can take 5 to 10 minutes, during which the patient will see some lights.[81]

Patient Safety

Clinicians and patients should be mindful of the following issues associated with OCTA to optimize patient safety and outcomes:

- **Noninvasive nature:** OCTA stands out as a noninvasive imaging modality, significantly reducing risks to patient safety compared to traditional fluorescein angiography or ICGA, which involves dye injections. The lack of dye minimizes the risk of allergic reactions, systemic toxicity, or discomfort, making OCTA a safer alternative, especially for patients with dye allergies or those at higher risk for systemic reactions.[82]
- **No radiation exposure:** OCTA uses light waves (near-infrared) to capture eye images, thus eliminating radiation exposure. This safety feature is particularly relevant for patients who require frequent imaging for chronic diseases like diabetic retinopathy or AMD, where cumulative radiation exposure can be a concern in other imaging techniques.[83]
- **Patient comfort:** OCTA is a quick and painless procedure. Patients are asked to rest their chin and forehead on supports during the imaging. While mild discomfort may be due to the bright light exposure, the procedure is generally well-tolerated. Ensuring patient comfort is vital to accurate imaging, as movement or poor cooperation may lead to image artifacts.[84]
- **Minimal contraindications:** OCTA has very few contraindications. Unlike traditional angiography, which may be contraindicated in patients with kidney issues due to the dye or in pregnancy, OCTA can be safely performed in almost any patient. The only limitation may be individuals with significant media opacities (eg, dense cataracts), which may interfere with image acquisition.[85]
- **Artifact management:** While OCTA is safe, patients should be informed that image quality can be affected by motion artifacts, poor fixation, or ocular surface abnormalities like dry eye. Eye care clinicians should emphasize the importance of cooperation during the scan to avoid false positives or inaccurate representations of blood flow.[70]
- **Clinical limitations:** OCTA is highly effective for visualizing blood vessels but does not detect leakage, which is often critical for diagnosing conditions like macular edema. Patients should be educated that OCTA might be used with other tests if leakage or dye-based information is required for accurate diagnosis.[66]

Patient Education

Clinicians should discuss the following aspects of the OCTA procedure with patients to ensure their complete understanding of this study:

- **Understanding the procedure:** Patients should be informed that OCTA is safe and noninvasive. Clinicians should also explain that the test is quick and typically takes 10 to 15 minutes without discomfort during or after the procedure. Patients should know that no injections, contact lenses, or contrast agents are required.[72]
- **Purpose of the test:** The purpose of the OCTA test should be clearly communicated to the patient. Patients should understand that OCTA allows doctors to visualize blood flow in the retina and choroid and assess for abnormalities in the blood vessels that could signify disease. For example, in diabetic retinopathy, OCTA can detect areas where the blood supply is compromised, and in AMD cases, OCTA can help detect new blood vessel growth. This can make a significant difference in early diagnosis and treatment.[40]
- **Follow-up testing:** Patients should be educated that OCTA may be repeated periodically to monitor their condition. For chronic diseases like glaucoma or diabetic retinopathy, ongoing monitoring of blood vessel changes may be required to evaluate disease progression and treatment efficacy. Patients should be aware of the non-cumulative nature of OCTA, meaning it can be performed as often as necessary without any risk of harm. [40]
- **Interpretation of results:** Patients may not always understand the implications of their OCTA results. Educating them on what the images show, such as areas of nonperfusion or abnormal vessel growth, helps patients better understand their condition. Explaining that OCTA can detect vascular changes before symptoms appear is essential, allowing for earlier and potentially more effective interventions.[16]
- **Managing expectations:** Patients should be informed that while OCTA provides valuable information, this study may not replace all other diagnostic tools. For example, traditional fluorescein angiography may still be required in cases where leakage is suspected. Clear communication about the complementary role of OCTA can manage patient expectations and reduce anxiety about needing additional tests.[86]
- **Postprocedure information:** Patients should be informed that no specific postprocedure care is required after the OCTA test. Patients do not need to worry about adverse effects like nausea, dizziness, or allergic reactions, as no dyes are used. They can resume their normal activities immediately following the procedure.[86]
- **Cost and accessibility:** Patients should be educated on OCTA's cost and insurance coverage. Although this study is increasingly becoming the standard of care in many practices, not all insurance plans may cover OCTA. Discussing the financial aspect upfront ensures patients are not surprised by out-of-pocket costs. However, emphasizing OCTA's noninvasive, dye-free benefits could underscore its value in comprehensive eye care.[87]
- **Management implications:** Patients should understand how OCTA could change the management of their disease. For example, OCTA may help detect changes in blood flow that suggest worsening disease before vision loss occurs, allowing for earlier intervention. For glaucoma patients, OCTA can track microvascular changes in the optic nerve, which could indicate the need for a change in treatment even when intraocular pressure appears stable.[2]

In summary, OCTA is a powerful and safe tool for diagnosing and monitoring eye diseases. By educating patients on the procedure's benefits, safety, and role in comprehensive eye care, clinicians can help patients feel more comfortable with the test, ensuring better cooperation and, ultimately, better outcomes. As OCTA technology advances, it will likely play an even greater role in eye disease management, making patient education a critical component of its success.

Clinical Significance

The method in which OCTA images are obtained allows for the tandem viewing of correlating en-face and cross-sectional B-scans, allowing for the evaluation of anatomical features with microvascular features visualized on OCTA. The clinical significance of OCTA continues to mature as technology usage grows. Conditions showing significant evidence for the utility of OCTA are glaucoma, uveitis, and various retinal pathologies. Diabetic retinopathy is associated with extensive effects on the retinal vasculature.[88] Identifying retinal changes, such as microaneurysms or neovascular complexes, and quantifying nonperfused areas of the eye (eg, the FAZ) using OCTA can aid in managing diabetic retinopathy.[89]

OCTA can also aid the evaluation of changes in choriocapillaris flow, analysis of choroidal neovascular membranes, and detection of retinal changes in age-related macular degeneration. Furthermore, using OCTA has potential benefits in managing other retinal diseases, such as central serous chorioretinopathy, macular telangiectasia, vascular occlusion, and choroidal neovascular membranes.[90][62][91][92][93][94][95][96][97]

OCTA is a noninvasive imaging modality that provides detailed retinal and choroidal microvasculature visualization. Since its introduction, OCTA has revolutionized ophthalmic imaging by allowing clinicians to observe vascular changes in various eye diseases without requiring dye injections, as is necessary with traditional fluorescein angiography and ICGA. OCTA's ability to capture high-resolution, depth-resolved images of blood flow within the retina and choroid makes it invaluable in diagnosing and managing various ocular conditions.[17]

Macular Degeneration

AMD is a leading cause of blindness in older adults. OCTA plays a critical role in detecting early changes in the choriocapillaris and identifying the presence of choroidal neovascularization, a hallmark of the "wet" form of AMD. The ability of OCTA to visualize CNV without dye injections offers a safer and more efficient method for monitoring disease progression and treatment response. In the "dry" form of AMD, OCTA can help track atrophic changes and the loss of the choriocapillaris, which may precede the development of geographic atrophy.[98]

Diabetic Retinopathy

OCTA has proven highly beneficial in managing diabetic retinopathy. It can detect microvascular abnormalities, such as capillary dropout, microaneurysms, and areas of non-perfusion, which are often the precursors to more severe diabetic eye disease. Visualizing the retinal vasculature in fine detail allows clinicians to assess the progression of DR more accurately. OCTA has been shown to detect macular ischemia and other changes before they become evident on standard fundus examinations or fluorescein angiography. In addition, OCTA can monitor the response to treatments such as anti-VEGF injections or laser therapy.[16]

Glaucoma

In glaucoma, OCTA is increasingly used to assess changes in the optic nerve head microvasculature and the peripapillary capillary network. Studies have demonstrated that reduced perfusion in the optic nerve head correlates with disease severity in glaucoma patients. OCTA can visualize these changes early, even before visual field defects are detectable. This makes OCTA a valuable tool in the early diagnosis of glaucoma, particularly in normotensive glaucoma, where structural changes may be subtle, and traditional methods may miss early signs of disease. OCTA also helps monitor the progression of the disease by tracking changes in the perfusion of the retinal nerve fiber layer and the ganglion cell complex.[43]

Retinal Vein and Artery Occlusions

OCTA is highly effective in diagnosing and monitoring RVO and retinal artery occlusions (RAO). In RVO, OCTA can visualize areas of capillary nonperfusion, vascular congestion, and collateral vessel formation. This imaging modality allows clinicians to assess the extent of ischemia and monitor the development of macular edema or neovascularization, both of which are critical for treatment decisions. In RAO, OCTA can help delineate the area of retinal ischemia, providing insights into the severity of the occlusion and guiding therapeutic interventions.[42]

Choroidal Diseases

Choroidal neovascularization associated with conditions like pathologic myopia, central serous chorioretinopathy (CSCR), and inflammatory choroidal diseases can be easily identified using OCTA. Unlike traditional dye-based angiography, OCTA can distinguish type 1 and type 2 CNV by analyzing different retinal layers, which is essential for diagnosis and treatment planning. In CSCR, OCTA can identify abnormal choroidal circulation and help track fluid dynamics within the retina and choroid.[99]

Retinal Dystrophies

In patients with inherited retinal dystrophies (eg, retinitis pigmentosa and Stargardt disease) OCTA can reveal abnormal vascular patterns and capillary dropout in the deeper retinal layers. This can provide valuable information on the progression of these diseases, especially since the structural changes often precede visual decline. OCTA may also help monitor responses to emerging gene and cell therapies to preserve or restore retinal function.[33]

Ocular Ischemic Syndromes

OCTA is also helpful in detecting ocular ischemic syndrome resulting from carotid artery stenosis. By visualizing reduced blood flow in both the superficial and deep retinal layers, OCTA can help diagnose this condition earlier, potentially guiding timely intervention to prevent further ischemic damage.[34]

Uveitis

In inflammatory diseases, such as uveitis, OCTA can provide insights into the involvement of the retinal vasculature. In cases where posterior uveitis affects the retina or choroid, OCTA can detect early signs of vasculitis, choroidal neovascularization, or macular edema, guiding treatment decisions more effectively than traditional imaging methods. [100]

Vascular Changes in Systemic Diseases

Systemic diseases (eg, hypertension and systemic lupus erythematosus) can affect the ocular vasculature. OCTA offers a noninvasive way to detect subtle changes in the retinal and choroidal microvasculature, which may reflect systemic disease severity. Monitoring these changes could provide clinicians with important clues regarding the progression of systemic conditions.[101]

Nonproliferative and Proliferative Diseases

A key advantage of OCTA is its ability to differentiate between nonproliferative and proliferative vascular diseases. For example, in diabetic retinopathy, OCTA can identify areas of neovascularization indicative of progression to the proliferative stage, allowing timely intervention. Similarly, in AMD, the development of CNV can be detected early, even before significant leakage occurs, facilitating early anti-VEGF treatment.[40]

Limitations of Optical Coherence Tomography Angiography

While OCTA offers numerous advantages, it has some limitations. One key limitation is its inability to detect leakage, a feature often visualized in dye-based angiography. For conditions where leakage is crucial for diagnosis and treatment planning, such as diabetic macular edema, fluorescein angiography may still be necessary. Additionally, artifacts can affect OCTA images due to patient movement, poor fixation, or media opacities like cataracts.[39]

Consequently, OCTA has emerged as a powerful tool in diagnosing, monitoring, and managing various retinal and choroidal diseases. Its noninvasive nature and ability to provide detailed, depth-resolved microvasculature images make it particularly valuable for early disease detection and long-term monitoring. As OCTA technology continues to evolve, it will likely play an even more significant role in routine clinical practice and research, further improving patient care in ophthalmology.[39]

Review Questions

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